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ADJUSTABLE FIXATION DEVICE

Abstract

An adjustable fixation device includes a cleat portion and a suture portion, where the suture portion has a loop portion and a suture tail. Pulling on the suture tail places in tension a graft coupled to the loop portion. The suture is arranged to convey a mechanical advantage so that substantial force can be readily and controllably applied to the graft.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 16/207,652 filed Dec. 3, 2018, which is a bypass continuation-in-part of International application no. PCT/US2017/035855 filed on Jun. 2, 2017, which claims the benefit of U.S. provisional patent application No. 62/344,510 filed on Jun. 2, 2016, and claims the benefit of U.S. provisional patent application No. 62/405,912 filed on Oct. 8, 2016, and this application also claims the benefit of U.S. provisional patent application No. 62/612,518 filed on Dec. 31, 2017, the disclosures of which are hereby incorporated by reference in their entireties.

FIELD OF THE INVENTION

[0002] The present invention relates to adjustable fixation devices and more particularly to an adjustable graft fixation devices for surgical applications.

BACKGROUND OF THE INVENTION

[0003] The ability to fix tissue effectively is central to the surgical process. In surgery, and in particular in arthroscopic surgery, fixation of tissue must be achieved under constraints of limited access. It is understood by those of skill in the art that fixation failure is a particularly important failure mode in the early postoperative period, and although many fixation approaches have been developed, the need for continued improvement remains widely acknowledged.

[0004] While a variety of techniques have been developed for coupling grans of soft tissue to bone, it remains difficult to achieve good results in certain particularly demanding procedures. For example, the repair of an anterior cruciate ligament is a surgical procedure where the parameters of graft tension and graft positioning have a strong bearing on surgical results.

SUMMARY OF THE INVENTION

[0005] An adjustable fixation device includes a cleat comprising first and second cleat openings and a connector seat between the openings. A flexible connector can have first and second cleat loops through the first and second openings of the cleat and a tensioning end portion. The connector comprises an intra-connector sliding loop engagement portion and a connector-device sliding loop engagement portion. The tensioning end portion of the flexible connector further comprises an axial sliding connector-connector engagement portion and a transverse sliding connector-connector engagement portion transversely engaging the first and second cleat loops and extending between portions of the cleat loops and the connector seat. The flexible connector can be a suture.

[0006] The axial sliding connector-connector engagement portion can include a portion of the tensioning end being positioned in an axial channel through a portion of a cleat loop. The axial channel can be formed by axial threading of the tensioning end through the portion of the cleat loop.

[0007] The transverse sliding connector-connector engagement portion can include a portion of the tensioning end being transversely threaded through a cleat loop. The transverse sliding connector-

connector engagement portion can include a portion of the tensioning end being positioned in a transverse channel through a portion of a cleat loop. The transverse channel can be formed by transverse threading of the tensioning end through the connector-connector engagement portion of the cleat loop.

[0008] At least one of said first and second cleat openings can be tapered. The cleat can further include lateral guide holes for receiving guiding sutures.

[0009] A protective cover can be provided for the flexible connector at a portion of the flexible connector comprising at least one selected from the group consisting of the intra-connector sliding loop engagement portion and the connector-device sliding loop engagement portion. The protective cover reduces friction and chafing.

[0010] An adjustable fixation system includes an adjustable fixation device. The adjustable fixation device can include a cleat comprising first and second cleat openings and a connector seat between the openings. A flexible connector having first and second cleat loops through the first and second openings of the cleat and has a tensioning end portion. The connector can include an intra-connector sliding loop engagement portion and a connector-device sliding loop engagement portion. The tensioning end portion of the flexible connector further can include an axial sliding connector-connector engagement portion and a transverse sliding connector-connector engagement portion transversely engaging the first and second cleat loops and extending between portions of the cleat loops and the connector seat. An implantable medical device is slidably connected to the adjustable fixation device at the connector-device sliding loop engagement portion.

[0011] The implantable device can be a graft. The implantable medical device can be a second cleat comprising first and second holes. The connector-device engagement portion of the flexible connector can be threaded through one of the holes in a first direction and through the other of the holes in a second direction forming a sliding connector-device engagement. The implantable medical device can include a plate comprising first and second holes. The connector-device engagement portion of the flexible connector can be threaded through one of the holes in a first direction and through the other of the holes in a second direction forming a sliding connector-device engagement.

[0012] The adjustable fixation system can further include a second adjustable fixation device having a second connector-device sliding loop engagement portion. The implantable medical device can be slidably connected between the connector-device sliding loop engagement portion of the adjustable fixation device and the connector-device sliding loop engagement portion of the second adjustable fixation device.

[0013] A method for the adjustable fixation of an implantable medical device can include the step of providing an adjustable fixation device which includes a cleat comprising first and second cleat openings and a connector seat between the openings, and a flexible connector. The flexible connector can have first and second cleat loops through the first and second openings of the cleat and a tensioning end portion. The connector can include an intra-connector sliding loop engagement portion and a connector-device sliding loop engagement portion. The tensioning end portion of the flexible connector can further include axial sliding connector-connector engagement portion and a transverse sliding connector-connector engagement portion transversely engaging the first and second cleat loops and extending between portions of the cleat loops and the connector seat.

[0014] The method also includes slidably connecting an implantable medical device to the adjustable fixation device at the connector-device sliding loop engagement portion. The adjustable fixation device and the implantable medical device are positioned in the body of the patient. A tensioning force is applied to the tensioning end portion of the flexible connector to tighten and secure the adjustable fixation device and the implantable medical device in position in the body of the patient.

[0015] The flexible connector can be a suture. The axial sliding connector-connector engagement

portion can include a portion of the tensioning end being positioned in an axial channel through a portion of a cleat loop. The axial channel can be formed by axial threading of the tensioning end through the portion of the cleat loop. The transverse sliding connector-connector engagement portion can include a portion of the tensioning end being transversely threaded through a cleat loop. The transverse sliding connector-connector engagement portion can include a portion of the tensioning end being positioned in a transverse channel through a portion of a cleat loop. The transverse channel can be formed by transverse threading of the tensioning and through the connector-connector engagement portion of the cleat loop.

[0016] The method can be used for a number of different surgical repair procedures. These include, but are not limited to, anterior cruciate ligament (ACL) repair, syndesmosis repair and ligamentum teres repair.

[0017] An adjustable fixation device can include a cleat, said cleat including a body member, the body member having an upper surface and a lower surface, the upper surface being disposed in substantially parallel spaced relation to the lower surface, the cleat having disposed therethrough between the upper and lower surfaces a first symmetric hole and a second tapered hole; and a suture portion, wherein a first region of the suture portion is disposed within the first symmetric hole and a second region of the suture portion is disposed within the tapered hole, the suture being configured to include an adjustable splice. The suture portion can further include a loop portion. The loop portion can be disposed coaxially within a sheath. The sheath can include first and second tails for fixing a graft adjacent to the loop portion of said suture portion.

[0018] An adjustable fixation device can include a cleat portion and a suture portion, the suture portion including a suture tail, the suture tail being configured, when placed in tension, to tighten a loop of the suture portion so as to convey a mechanical advantage from the tail portion to the loop. The adjustable fixation device can provide a 4:1 mechanical advantage. The loop portion can be disposed on a first side of the cleat portion and the tail portion can be disposed on a second opposite side of the cleat portion, whereby the tail may be placed in tension after the cleat portion is installed to support a surgical graft with the loop portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] There are shown in the drawings embodiments that are presently preferred it being understood that the invention is not limited to the arrangements and instrumentalities shown, wherein:

[0020] FIG. 1 shows, in schematic view, an exemplary adjustable fixation device prepared according to principles of the invention;

[0021] FIG. 2 shows, in schematic perspective view, an exemplary cleat portion of an exemplary adjustable fixation device prepared according to principles of the invention;

[0022] FIG. 3A shows, in schematic cross-section, portions of an adjustable fixation device including a cleat and a suture according to principles of the invention;

[0023] FIG. 3B shows a further aspect of an adjustable fixation device including a cleat and a suture according to principles of the invention;

[0024] FIG. 30 shows a still further aspect of an adjustable fixation device including a cleat and a suture according to principles of the invention; and

[0025] FIGS. 4-25 show, in schematic perspective view, selected steps in an exemplary method for preparation of an adjustable fixation device including a cleat and a suture according to principles of the invention.

[0026] FIG. 26 is a side elevation partially in cross-section and partially in phantom, of an adjustable fixation device in a first stage of assembly.

[0027] FIG. **27** is a side elevation partially in cross-section of an adjustable fixation device in a second stage of assembly.

[0028] FIG. **28** is a side elevation partially in cross-section of an adjustable fixation device in a third stage of assembly.

[0029] FIG. **29** is a side elevation partially in cross-section and partially in phantom, of an adjustable fixation device in a fourth stage of assembly.

[0030] FIG. **30** is a side elevation partially in cross-section and partially in phantom, of an adjustable fixation device in a fifth stage of assembly.

[0031] FIG. **31** is a side elevation of an adjustable fixation device and a surgical graft.

[0032] FIG. **32** is a schematic perspective, partially broken away, of an adjustable fixation device according to the invention and a surgical graft being inserted into the femur and tibia in a method according to the invention.

[0033] FIG. **33** is a schematic perspective, partially broken away, of an adjustable fixation device and a surgical graft being inserted into the femur and tibia in a subsequent stage of the method.

[0034] FIG. **34** is a schematic perspective, partially broken away, of an adjustable fixation device in a surgical graft being inserted into the femur and tibia in a stage of the method subsequent to that shown in FIG. **33**.

[0035] FIG. **35** is a schematic side elevation, partially in cross section, of an adjustable fixation device and a surgical graft in an initial untightened position.

[0036] FIG. **36** is a plan view, partially in cross-section, of a cleat and flexible connector of the adjustable fixation device in the initial untightened position.

[0037] FIG. **36A** is an expanded view of area FIG. **36A** in FIG. **36**.

[0038] FIG. **37** is a schematic side elevation, partially in cross section, of an adjustable fixation device and a surgical graft in a tightened and secured position.

[0039] FIG. **38** is a plan view, partially in cross-section, of a cleat and a flexible connector of an adjustable fixation device in a tightened and secured position.

[0040] FIG. **39** is a schematic side elevation of an adjustable fixation device and a surgical graft in a tightened and secured position in a femur.

[0041] FIG. **40** is a side perspective view of a double-sided embodiment of an adjustable fixation device with a surgical graft.

[0042] FIG. **41** is a schematic perspective view, partially in phantom, showing an initial step in the insertion of the double-sided embodiment in a tibia and femur.

[0043] FIG. **42** is a schematic perspective view, partially broken away, showing the completion of the installation of the double-sided embodiment in the tibia and femur.

[0044] FIG. **43** is a top perspective view of an alternative embodiment of an adjustable fixation device suitable for syndesmosis repair.

[0045] FIG. **44** is a bottom perspective view of the adjustable fixation device of FIG. **43**.

[0046] FIG. **45** is a perspective view of the adjustable fixation device for syndesmosis repair of FIG. **43** secured to the tibia and to the fibula,

[0047] FIG. **46** is a side elevation, partially broken away, of the adjustable fixation device for syndesmosis repair of FIG. **43** secured to the tibia and to the fibula.

DETAILED DESCRIPTION OF THE INVENTION

[0048] In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the present invention. It will be apparent, however, to one skilled in the art that the present invention may be practiced without these specific details. It should be noted that, while the various figures show respective aspects of the invention, no one figure is intended to show the entire invention. Rather, the figures together illustrate the invention in its various aspects and principles. As such, it should not be presumed that any particular figure is exclusively related to a discrete aspect or species of the invention. To the contrary, one of skill in the art would appreciate that the figures taken together reflect various

embodiments exemplifying the invention.

[0049] Referenced throughout the specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, the appearance of the phrases “in one embodiment” or “In an embodiment” in various places throughout the specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. Directional references such as upward and downward are provided for purposes of a reference for description, and it should be understood that the invention can be oriented in any direction within the body.

[0050] For clarity of disclosure, the terms “flat oval” and “tapered flat oval” are now defined. The term “flat oval” is intended to describe a geometric shape including two semicircles of equal radius joined at their open ends by first and second line segments of equal length. The term tapered flat oval is intended to describe a geometric shape including two approximate semicircles of unequal radius joined at their open ends by first and second line segments of equal length.

[0051] An adjustable fixation device according to the invention can include a cleat comprising first and second cleat openings and a connector seat between the openings. A flexible connector can form first and second cleat loops through the first and second openings of the cleat and a tensioning end portion. The connector can include an intra-connector sliding loop engagement portion and a connector-device sliding loop engagement portion. The tensioning end portion of the flexible connector can further comprise an axial sliding connector-connector engagement portion and a transverse sliding connector-connector engagement portion transversely engaging the first and second cleat loops and extending between portions of the cleat loops and the connector seat.

[0052] The flexible connector can be of many suitable constructions and materials. The flexible connector can be braided or single filament. The dimensions of the flexible connector can vary depending on the repair that is being performed and the strength and dimensions necessary for the repair. The flexible connector can be made from any suitable flexible medical grade material, including polymeric materials and flexible metal materials. The flexible connector can be a suture, and in this application the terms suture and flexible connector will sometimes be used interchangeably.

[0053] The axial sliding connector-connector engagement portion can have a portion of the tensioning end positioned in an axial channel through a portion of a cleat loop. The axial channel can be formed by axial threading of the tensioning end through the portion of the cleat loop. The transverse sliding connector-connector engagement portion can include a portion of the tensioning end being transversely threaded through a cleat loop. The transverse sliding connector-connector engagement portion can include a portion of the tensioning end being positioned in a transverse channel through a portion of a cleat loop.

[0054] The transverse channel can be formed by transverse threading of the tensioning end through the connector-connector engagement portion of the cleat loop. The cleat can further include lateral guide holes for receiving guiding sutures for use in positioning the adjustable fixation device at the target location.

[0055] An implantable medical device is slidably connected to the adjustable fixation device at the connector-device sliding loop engagement portion. The implantable medical device can be selected from many different implantable medical devices. One such implantable device is a graft. The implantable medical device can be a second cleat. The second cleat can have for example first and second holes. The connector-device engagement portion of the flexible connector can be threaded through one of the holes in a first direction and through the other of the holes in a second direction forming a sliding connector-device engagement.

[0056] The implantable medical device can include a second cleat, plate or other structure comprising first and second holes, and wherein the connector-device engagement portion of the flexible connector is threaded through one of the holes in a first direction and through the other of

the holes in a second direction forming a sliding connector-device engagement. Such a device can be used for example in syndesmosis repair procedures.

[0057] The adjustable fixation system can include a second adjustable fixation device having a second connector-device sliding loop engagement portion. The implantable medical device can be slidably connected between the connector-device sliding loop engagement portion of the adjustable fixation device and the connector device sliding loop engagement portion of the second adjustable fixation device.

[0058] Modifications are possible, A covering portion can be provided over the flexible connector or the graft at the sliding engagement portions to minimize sliding friction or chafing from sliding friction. The cover can be a piece of protective suture through which the flexible connector portions are threaded, an external cover applied over these portions of the flexible connector, or a low friction protective coating composition applied to the flexible connector.

[0059] The axial and transverse sliding engagement of the sliding engagement portions of the flexible connector are shown as being formed by threading one portion of the flexible connector through another, axially or transversely. Such a sliding connection can also be accomplished by one or more fittings with channel portions through which the sliding connection can be accomplished. Such channel portions can be oriented axially or transversely as necessary. Such fittings can be provided in-line or crimped or otherwise attached to the flexible connector at the proper position.

[0060] A method for the adjustable fixation of an implantable medical device can include positioning the adjustable fixation device using guide sutures or some other placement apparatus such as forceps. The cleat is positioned against bone or some other body part that will provide resistance as tension is applied to the tensioning end portion. The opposing end of the implantable medical device is similarly secured. The implantable medical device can be slidably connected to the adjustable fixation device either before or after positioning of the adjustable fixation device at the target location. A tensioning force is then applied to the tensioning end portion of the flexible connector to tighten and secure the adjustable fixation device and the implantable medical device in position in the body of the patient.

[0061] The method of the invention can be used to perform a number of surgical procedures. These include anterior cruciate ligament (ACL) repair, and syndesmosis repair where the implantable medical device is a plate for engaging and supporting the fibula. The invention is useful for ligamentum teres reconstruction procedures, and for other procedures.

[0062] For example, in anterior cruciate ligament (ACL) repair, as illustrated schematically **100** in FIG. **1**, a damaged ACL is removed and replaced by graft tissue **102**. The graft tissue is disposed within first **104** and second **106** channels prepared in the tibia **105** and femur **107** respectively. The graft tissue **102** is, in the illustrated repair, coupled to the tibia **105** with an interference fixation where a helically threaded bone anchor **108** mechanically compresses a portion **110** of the graft tissue **102** between an external circumferential surface of the bone anchor **108** and an internal surface region **112** of the channel **104**.

[0063] As shown, the graft tissue **102** is coupled to the femur **107** with a combination of a cleat **114** and a suture **116**. The cleat **114** is disposed across an aperture **118** where the second channel **106** exits the femur. The cleat, sometimes referred to as a button, is substantially rigid and supports the suture **116** in tension. A proximal surface region **120** of the cleat is disposed in contact with corresponding surface regions of the bone surrounding aperture **118**.

[0064] In various surgical repairs, and particularly in ACL repair, the forces borne by the graft are relatively large. Moreover, while it is desirable to stabilize the joint to allow healing. It is also important to maintain activity in a joint to avoid loss of range of motion. Consequently, it is important that the position and tension applied to the graft **102** is correct. Disadvantages of existing methods and apparatus include the difficulty involved in achieving a reliable coupling between soft tissue and bone, and ensuring that the ultimate graft assembly has both the right length and the right tension.

[0065] Through personal experience, analysis, and careful observation, the present inventors have come to understand that the effectiveness of tissue coupling in a wide variety of surgical procedures can be improved by providing a fixturing device that permits ready adjustment of the position and tension of soft tissue elements secured by the device. In response, the inventors have conceived and reduced to practice and adjustable surgical cleat and suture combination. Accordingly, the present invention includes an adjustable surgical cleat, and a combination of adjustable surgical cleat and suture.

[0066] According to the present invention, a combination of the splice and cleat design work together to provide a high breaking strength. The cleat features narrow, elongated openings designed to stack the suture onto itself to maintain consistency, and can include a v-shaped notch in one of the grooves designed to fasten the adjustment tail. The v-shape is specifically designed to clamp the tail without the need for a hard edge. This allows for a larger radius inside the cleat, which greatly reduces shearing between the suture and cleat. Also improving the product's breaking strength is the splice design itself. The design creates a 4:1 mechanical advantage at the adjustment tail and effectively distributes the load over more lengths of suture than comparable products.

[0067] Minimal elongation is achieved by combining the adjustable loop splice with the mechanical fixation of the tail. An inherent problem with alternative technologies and that of the loop splice itself is the slip of the tail back into the splice during cyclic loading. According to the present invention, to prevent 'tail slip', the tail is spliced beneath both lengths of suture at the cleat. This utilizes the tension on the loop to clamp the tail into the cleat's v shaped groove which nearly eliminates tail slip.

[0068] Another matter of elongation to be minimized is the initial displacement. As a product of the splice and suture weave tightening after an adjustment in length, the 4:1 mechanical advantage allows only a quarter of the splice's initial elongation to be translated into graft displacement. This is a tremendous advantage over earlier approaches, many of which translate initial splice elongation directly into graft displacement.

[0069] A great benefit of a 4:1 mechanical advantage splice design is its adjustability. It allows surgeons to apply tension to the graft far more easily than with competing products. Not only can tension be applied more easily, but greater tension can be applied, allowing surgeons improved adjustability when fixing a graft to bone. Furthermore, the single tail simplifies adjustment, and allows the surgeon a free hand while applying tension to the graft.

[0070] FIG. 2 shows, in schematic perspective view, an exemplary cleat **200** prepared according to principles of the invention. The cleat includes a body member **202** with an outer surface **204**. An inner surface **206** is disposed in substantially parallel spaced relation to the outer surface **204**.

[0071] In the illustrated embodiment, the body member **202** substantially exhibits mirror symmetry across a centerline **208** of the outer surface **204**. It should be understood, however, that a wide variety of other shapes will be employed in alternative embodiments of the invention.

[0072] An internal circumferential surface **210** defines a bore **212** through the body member **202**, between the outer surface **204** and the inner surface **206**, generally proximate to a first end **214** of the body member. In the illustrated embodiment, the internal circumferential surface **210** defines respective generally circular apertures, e.g. **216**, wherein meets the outer **204** and inner **206** surfaces. A corresponding internal surface **218** defines a similar bore **220** and similar apertures, e.g. **222** at a second opposite end **224** of the body member **202**.

[0073] One of skill in the art will understand that alternative shapes of aperture **216**, **222** are contemplated to be within the scope of the invention. Thus, the apertures may be polygonal, elliptical, stellate, randomly shape, etc. according to corresponding abundance of the invention. Moreover the apertures of first and second ends of the body member **202** may be differently shaped. Additionally, in certain embodiments one or more of the two apertures will be omitted, or additional apertures may be added that are not illustrated by the exemplary cleat **200**.

[0074] As in the illustrated embodiment, in certain embodiments edges defining the apertures, e.g.,

226, 228 will be chamfered or curved, or otherwise softened so as to reduce the likelihood of a suture being abraded or cut by the edge.

[0075] Two further internal surface regions **230, 232** define respective third **234** and fourth **236** bores through the body member **202**, with corresponding apertures, e.g. **238, 240** at the outer **204** and inner surfaces **206**. Like edges **226, 228**, edges of apertures **238, 240**, will be beveled, chamfered or curved in various embodiments of the invention.

[0076] Aperture **238** is substantially flat oval in shape. Aperture **240** is tapered flat oval in shape, and has a relatively smaller radius at its inner curve **242** as compared with its outer curve **244**. Both the flat oval aperture **238** and the tapered flat oval aperture **240** have respective longitudinal axes. In the illustrated embodiment, these axes are substantially parallel to the centerline **208** of the outer surface **204** of the body member **202**. It should be understood, however, that the longitudinal axes of the apertures **234, 236**, and their orientation more generally with respect to the body member **202** will be different in different embodiments of the invention.

[0077] A recess or groove **242** is disposed between the proximate ends of aperture **234** and aperture **236**. Groove **242** is generally concave, as viewed from the perspective of FIG. 2, and in certain embodiments, has edges that are chamfered or curved. As will be understood upon review of the further description below, apertures **234, 236**, and groove **242** are sized to accommodate suture material for an adjustable cleat and suture combination.

[0078] In certain embodiments of the invention, body member **202** will include a titanium material. In other embodiments of the invention, body member **202** will include a PolyEther Ether Keystone (PEEK) material. In various embodiments of the invention, the material of body member **202** will include a bio-absorbable material and in other embodiments of the invention the body member **202** will include a non-bioabsorbable material. Any of a wide variety of biocompatible materials will be used in corresponding embodiments of the invention.

[0079] FIG. 3A shows, in schematic cross-section, portions of an adjustable fixation device **300** including a cleat **302** and a suture **340**. The illustrated cleat, **302** is similar to cleat **200** of FIG. 2 and includes a body member **304**, an outer surface region **306** and an inner surface region **308** disposed in substantially parallel spaced relation to one another. First **310** and second **312** outer bores connect the surface region **306, 308** each in relative to proximity to a respective end **314, 316** of the body member **304**.

[0080] Also disposed between outer surface region **306** and inner surface region **308** are a third generally tapered flat oval bore **318** and a fourth generally flat oval bore **320**.

[0081] As shown in cutaway view, a braided suture **340** has a first end **342** captured longitudinally within a portion **344** of the same suture. Proceeding from the captured end **342**, the suture proceeds outwardly through an aperture **346** in a wall of the suture **340**. Thereafter, a portion of the suture **348** is disposed within the tapered flat oval bore **318** of cleat **302** and passes **350** above a groove **352** of the cleat **302**. Proceeding further along the suture **340** in the same direction, a portion **354** of the suture is disposed within flat oval aperture **320**. Proceeding in the same direction, one passes again the aperture **346** and proceeds to a portion of the suture disposed in an upwardly facing U-shaped bend **356**.

[0082] Proceeding thereafter in the same direction, a further portion **358** of the suture is disposed, once again, within flat-oval aperture **320** before a still further portion of the suture **360** is disposed once again above groove **352**. Proceeding thereafter in the same direction, another portion of the suture **362** is disposed again within the tapered flat-oval aperture **318**. Below this, is disposed a further aperture **364** in the wall of suture **340**.

[0083] Proceeding in the same direction, and as will be described shortly, a portion **366** of the suture has a further portion disposed coaxially therewithin. Thereafter the suture proceeds, in the same direction, past a further aperture **368** to a further upwardly facing U-shaped bend portion of the suture **370**.

[0084] Proceeding in the same direction, one arrives at a portion of the suture **372** disposed in a

downwardly facing U-shaped bend intercoupled with upwardly facing U-shaped portion **356**. Proceeding again in the same direction, an additional portion of the suture **374** forms a further upwardly facing U-shaped bend. Again, proceeding in the same direction, a portion **376** of the suture is disposed within aperture **368**.

[0085] The next proceeding portion of the suture **378** is disposed coaxially within suture portion **366** and thereafter exits through aperture **364**. Proceeding again in the same direction another portion **380** of suture **340** is disposed again within the tapered flat oval aperture **318** thereafter moving beyond the cleat **304** to pass into and immediately out of suture portion **350** transversely through respective apertures **382**, **384**, emerging as suture portion **386**. Thereafter, suture **340** passes upwardly to pass into and immediately out of suture portion **360** transversely through aperture **388**, **390** respectively, emerging as suture portion **392** which then forms a tail **394** of suture **340**.

[0086] As will be further discussed below, pulling **396** on tail **394** tends to urge suture portion **370** and **374** upwardly **398** toward surface **308** with a substantial (four times) mechanical advantage. Accordingly, and again as will be further discussed below, coupling a graft to suture portions **370** and **374** allows positional adjustment and tensioning of the graft by pulling on tail **394** of suture **340**.

[0087] Upon careful examination of FIG. **3A**, it will be apparent to the practitioner of skill in the art that, upon tensioning of the suture by pulling **396** on tail **394**, reactive forces **400** exerted by the graft will tend to pull suture portions **350** and **360** into groove **352** where they will tend to compress the suture portions passing through apertures **382**, **384** and **388**, **390**, thereby preventing motion in the direction of the reactive forces **400**. Likewise, constrictive forces applied to suture portion **378** by internal surface regions of suture portion **366** will tend to prevent such motion. Accordingly, a unidirectional apparatus is provided which allows tightening by pulling on tail **394** but which resists loosening in response to reactive forces **400**.

[0088] FIG. **3B** shows a further aspect of the invention in which suture portions **370** and **374** of fixation device **300**, described above, are disposed within a covering suture portion **402**. The covering suture portion serves to increase the surface area supporting a graft and reduce the opportunity for graft abrasion by suture portion **370** and **374**.

[0089] As will be apparent upon examination of the figure, and proceeding in the direction previously established suture portion **370** of suture **340** passes inward through an aperture **404** of cover suture portion **402** and proceeds generally coaxial to portion **402** until exiting from cover portion **402** through aperture **406** thereafter, suture **340** proceeds through previously-identified inverted U-shaped bend **372** and thereafter, as suture portion **374** reenters aperture **406**. Suture **340** then proceeds substantially coaxially within suture portion **402** reemerging from aperture **404** as suture portion **376**.

[0090] Apertures **404** and **406** are disposed medially within suture portion **402**, such that the extremities of suture portion **402** form respective tails **408**, **410**. As will be further discussed in relation to FIG. **4C**, tails **408** and **410** are beneficially applied in securing a graft to the assembly of suture portions **370**, **374** and **402**.

[0091] FIG. **3C** shows a further aspect of the invention in which the tails **408** and **410** of fixation device **300** are secured about a graft **412**, with a knot **414** thereby securing the graft **412** to the fixation device **300**. Once thus secured, and consistent with the discussion above, position and tension of the graft with respect to cleat **302** and along axis **416** can be adjusted by pulling **396** on tail **394** of suture **340**.

[0092] FIGS. **4-26** show, in schematic perspective view, selected steps in an exemplary method for preparation of one embodiment of a fixation device **500** according to principles of the invention. Prior to beginning assembly, a length of suture is marked and precut as follows. A temporary holding cleat is fastened to one end of a length of suture. This holding cleat facilitates construction and will be removed upon completion of assembly.

[0093] FIG. 4 shows a cleat 502 bearing a general similarity to cleat 302 described above. The cleat 502 includes a body member 504 with an outer surface 506, an inner surface 508, a tapered flat oval aperture 510 and a flat oval aperture 512. A length of suture 514 is disposed from an inner side 508 toward the outer side 506 through the tapered flat oval aperture 510, forming a U-shaped bend 516 and passing through the flat oval aperture 512 to return to the inner side 508 of the cleat 502. Defining the two ends of the suture 514 as first end 518 and second end 520, six marks are placed on the suture as follows: [0094] Mark 1—cut at this point to ensure proper length-of-taper in the eye-splice [0095] Mark 2—the second cross of the eye-splice [0096] Mark 3—the first cross of the eye-splice, locks in conjunction with Mark 2 [0097] Mark 4—used to align the suture so as to ensure proper loop length [0098] Mark 5—“splice in” where a tail enters the adjustable splice (below “splice out”) [0099] Mark 6—“splice out” where the tail exits the adjustable splice [0100] In certain embodiments of the invention, Mark 1 will be placed at 0 mm, Mark 2 at 19 mm, Mark 3 at 44 mm, Mark 4 at 70 mm, Mark 5 at 78 mm, Mark 6 at 92 mm. In other exemplary embodiments of the invention, Mark 1 will be placed at 0 mm, Mark 2 will be placed at 15 mm, Mark 3 will be placed at 36 mm, Mark 4 will be placed at 56 mm, Mark 5 will be placed at 58 mm, Mark 6 will be placed at 71 mm. One of skill in the art will appreciate that these dimensions are merely exemplary of a wide variety of different measurements and ratios that will be employed in respective embodiments according to the requirements of particular circumstances.

[0101] Referring again to FIG. 4 one sees on suture 514 the placement of Mark 1 522, Mark 2 524, Mark 3 526, Mark 4 528, Mark 5 530, and Mark 6 532. In practicing the method, one pulls the marked end of the suture through the tapered flat oval aperture (hereinafter tapered hole) 510 first and thereafter through the flat oval aperture (hereinafter symmetric hole) 512. The cleat is then centered between Mark 2 524 and Mark 3 526, and the marked side of the suture is facing outward 527.

[0102] FIG. 5 shows the preparation of fixation device 500 and the use of a suture passer 531 disposed through an aperture at Mark 3 526 and coupled at a distal end thereof to suture 514 at Mark 2 524. Thereafter, the suture passer 531 is withdrawn to bring the suture end through and place the device 500 into the state of FIG. 6.

[0103] FIG. 6 shows the resulting configuration in which a suture portion 532 is disposed within and transverse to suture portion 534, having passed through an aperture at Mark 3 526. The location of Mark 2 524 is visible.

[0104] FIG. 7 shows a further step of the method in which suture passer 531 is disposed through an aperture at Mark 2 524 and coupled to suture 514 as shown. The suture passer 531 is then withdrawn to bring suture end 518 through the aperture at Mark 2 524.

[0105] The result is as shown in FIG. 8. The marked side of the suture should continue to face outward 527. Thereafter, the cleat is moved in direction 527 while holding end 518 of the suture 514. This tends to bring the two apertures 526 and 524 towards each other and cinch the splice in place.

[0106] FIG. 9 shows the configuration of the cleat 502 and suture 514 once the splice has been cinched at 536.

[0107] As shown in FIG. 10 the suture passer 531 is then inserted into suture 514 at Mark 4 528, passed axially within the suture and out of the suture as close as possible to location 536. The suture passer 531 is then coupled to end 520 of suture 514 and withdrawn, bringing end 520 of the suture back through location 536 and out at Mark 4 528.

[0108] FIG. 11 shows a result with end 520 of suture 514 having been pulled axially through the splice at 538. The balance of suture 514 towards end 520 is then cut off in close proximity to where it emerges at 528. A mark is made 529 at the center of the eye splice loop on the suture facing the upper surface 506 of the cleat 502.

[0109] The result, as shown in FIG. 12 is a single end 518 of the suture 514, coupled with an eye-splice loop 540 to the cleat 502. At this point, a quality check will ensure that the suture is not

frayed, that the eye-splice ends are substantially coincident with Mark 2 524 and with Mark 3 526, that the transverse passages of the suture through the suture body are centered and symmetrical, and that the suture is not unduly twisted. In addition, it should be confirmed that the coaxial internal portion of the suture, previously coupled to end 520, terminates short of Mark 4 528 and that the tail of this coaxial internal portion is completely within the outer suture.

[0110] In a further step of the method, suture passer 531 is disposed through the symmetric hole 512 from the upper surface 506 towards a lower surface 508 of the cleat 502. End 518 of suture 514 is coupled to a distal end of the suture passer 531. The suture passer 531 is withdrawn bringing end 518 of suture 514 through the upper surface of the cleat 502. Thereafter, and as shown in FIG. 13, suture passer 531 is passed through tapered hole 510 of cleat 502 from the lower surface 508 towards the upper surface 506. A distal end of the suture passer 531 is coupled to suture 514 near its end 518. The suture passer is then withdrawn 542, bringing end 518 through tapered hole 510 and out on the lower surface 508 side of the cleat 502. This forms a loop 544 of suture 514 at the lower surface 508 of cleat 502 and a further loop 546 of suture 514 at the upper surface 506 of cleat 502.

[0111] FIG. 14 illustrates the method steps involved in preparing a sleeve for a fixation device 500 according to principles of the invention. A length of suture 550 is marked at symmetrically intermediate points 552, 554 to define a length of suture 556 therebetween. One of skill in the art will understand that length 556 is selected according to the requirements of a graft to be performed, and is related to a width of the graft tissue.

[0112] A point of a suture shuttle 553 is inserted at 554 and passed coaxially within the suture to 552, where the suture shuttle 553 point emerges from the suture. The suture shuttle is used to draw end 518 of suture 514 (as previously described above) through the suture 550.

[0113] FIG. 15 shows cleat 502 including suture loop 544 and suture loop 546 as discussed in relation to FIG. 13. Also visible is end 518 of suture 514 coupled to a loop 558 of the suture shuttle 553 at an intermediate point 560 of suture 514. The suture shuttle 553 is being used to draw end 518 of suture 514 through the suture (sleeve) 550 between points 552 and 554.

[0114] FIG. 16 shows a further step in the method of preparing a fixation device 500. As is evident from the figure, suture 514 is looped from the eye splice through loop 544 and passes through cleat 502 to form loop 546 at the upper surface of cleat 502. Thereafter it passes through cleat 502 again past the lower surface where it proceeds as suture portion 562, entering the sleeve at point 552 and exiting at point 554. Suture 514 then forms a further loop at portion 564, passing through loop 544 and thereafter passing through loop 558 of the suture shuttle 553. Thereafter, tension is applied 566 to a terminal cleat 568 of the suture shuttle, drawing end 518 of suture 514 back into sleeve 550 at point 554 and out again at point 552.

[0115] FIG. 17 shows a result, where portion 564 of suture 514 forms an interlinked loop 570 with loop 544.

[0116] FIG. 18 shows a further step in the method of forming a fixation device 500. Specifically, end 518 of suture 514 is drawn through tapered hole 510 from the inner side 508 towards the outer side 506 of cleat 502 with the suture passer (not shown). Tension is then applied to previously identified loop 546 to draw suture 514 through the cleat 502 and expand loop 546. The loop is expanded until Mark 4 528, Mark 5 530 and Mark 6 532 are at the upper surface 506 side of the cleat 502.

[0117] FIG. 19 shows a further step in the method of forming a fixation device 500. In this step, suture passer 531 is inserted into suture 514 at the opposite side of the suture from Mark 5 530, passed coaxially within the suture of loop 546, and passed out of suture 514 at Mark 6 532 on the same side of suture 514 as Mark 6. End 518 of suture 514 is coupled to the distal end of suture passer 531 and drawn back into the suture at Mark 6 532, through the suture 514 coaxially and back out of the suture at 530, thus forming an adjustable splice 574, as further shown in FIG. 20.

[0118] Referring further to FIG. 20, tension 578 is applied to suture 514 at region 580 and the

portion **582** of adjustable splice **574** is drawn through cleat **502** from upper surface **506** towards lower surface **508** so as to transfer portion **582** of the adjustable splice **574** to the lower surface side of the cleat.

[0119] This result is visible in FIG. **21** where end **518** of suture **514** emerges from the adjustable splice at Mark 6 **532** at the outer surface **506** side of cleat **502** while a further portion **582** of the adjustable splice **574** is disposed beyond the inner surface **508** of cleat **502**.

[0120] In certain embodiments, the method of preparing fixation device **500** now includes the steps of ensuring that no fraying of the sutures has happened, especially around the adjustable splice and sleeve, the assembly is inspected to ensure that the suture has run properly through the cleat. After the eye-splice loop **540**, the suture **514** should run into the symmetric hole **512** first, and thereafter into the tapered hole **510**. End **518** of suture **514** should exit the adjustable splice **574** on the same side of cleat **502** as upper surface **506**. The suture should enter the adjustable splice on the same side as, and directly through, Mark 6 **532** and exit the adjustable splice on the opposite side of the suture **514** from Mark 5 **530**. The length of the symmetric splice **574** should be equally divided on opposite sides of the cleat **502**.

[0121] FIG. **22** shows a further step in a method of forming an adjustable fixation device **500** including drawing the eye-splice loop **540** to the upper surface **506** side of cleat **502**, passing a needle **584** from an inner surface of the eye-splice loop through the mark at **529** made and described in relation to FIG. **11**. A distal end of the needle **584** is coupled to end **518** of suture **514** and drawn back through the suture from the outside to the inside of the eye-splice loop **540** at the mark **529**.

[0122] Thereafter, as shown in FIG. **23**, the needle **584** is employed to further draw the end **518** of the suture **514** through location **586** at the top of the eye-splice loop **542** and then through the suture **514** at Mark 4 **528**.

[0123] The adjustable fixation device **500** will now be in the configuration illustrated in FIG. **24**. The method that includes ensuring that there is no fraying of suture material, ensuring that crosses are properly through the center of the suture and all marks are facing away from the cleat.

[0124] Thereafter, to achieve the final configuration the eye splice is lightly pulled towards the lower side **508** of the cleat **502**. Thereafter, simultaneously pulling on the eye splice and end **518** of the suture **514** will further finalize positioning of the sutures. The suture is pulled towards the lower side **508** of the cleat **502** at the symmetric whole and that the entire loop is pulled from the bottom working gently to avoid punching. Finally it should be confirmed that the end of the suture **518** moves freely to adjust the size of the graft fixturing loop.

[0125] There is shown in FIGS. **26-30** an illustration of a sequence of steps to create an adjustable fixation device according to one embodiment of the invention. There is shown in FIG. **26** a cleat **600** having a bottom surface **601** and a top surface **602**, and an oval bore **604** and a tapered bore **608**. The cleat **600** can have circular apertures **610** and **612**. A bridge or seat portion **618** spans between the oval bore **604** and the tapered bore **608**.

[0126] A flexible connector **621** has a tail **622** and a portion of the flexible connector **621** passes through the oval bore **604** and over the bridge portion **618** and through the tapered bore **608** to form a downward facing inner bend **626**. A first end **630** of the flexible connector **621** is captured longitudinally within a portion **631** of the suture **621** and enters the flexible connector **621** at an aperture **634**. As shown in FIG. **27**, a downward facing outer bend **638** of the flexible connector **621** is returned to the oval bore **604** forming an upward facing bend **639**. A portion **646** of the flexible connector **621** passes through the tapered bore **608** such that the downward facing outer bend portion **638** is over the downward facing inner bend portion **626**.

[0127] As shown in FIGS. **28-29**, the tail **622** can be passed over the upward facing bend **639** to form a downward facing bend **642** that rests on the upward facing bend **639** with an ascending portion **648** and a descending portion **652**. An upward facing bend **650** is thereby formed. The descending portion **652** is turned upward at a bend **658** to form and an ascending portion **654**. A

portion **662** of the ascending portion **654** is captured longitudinally within a portion **661** of the portion **646** and enters at an aperture **666** and exits at an aperture **670**.

[0128] As shown in FIG. **30**, upon leaving the aperture **670** the flexible connector **621** transversely engages the downward facing inner bend **626** through an aperture **674** that is below a bottom surface **601** of the cleat **600**. A portion **675** of the flexible connector **621** leaving the aperture **674** abuts the bridge portion **618** and is positioned between the bridge portion **618** and a corresponding portion of the downward facing bend **626**. The portion **675** transversely enters a portion of the downward facing bend **626** above the seat portion **618** at an aperture **678**. The suture **621** then transversely passes through the downward facing outer bend **638** at an aperture **682**. The tail **622** extends above the cleat **600** to complete the adjustable fixation flexible connector portion **688** portion and with the cleat **600** the adjustable fixation device **690**. The tail **622** will provide tensioning to tension and secure the adjustable fixation device **690** in position.

[0129] There is shown in FIG. **31** a graft **700** in position with the adjustable fixation device **690**. The graft **700** can have securing sutures **701-704** at ends thereof. The adjustable fixation device **690** and the graft **700** together form a system similar to pulleys, as indicated by area A and B. In area A, the downward facing bend **642** slidably traverses over the upward facing bend **639** in the manner of a pulley. In area B, portion **684** of graft **700** slidably traverses over the upward facing bend **650** and the upward facing bend **658** in the manner of a pulley. Pulley system A provides a mechanical advantage of 2:1, and pulley system B provides a mechanical advantage of 2:1, so that together pulley system A and pulley system **8** provide a joint mechanical advantage for the adjustable fixation device **690** of 4:1.

[0130] The operation of the adjustable fixation device **690** is shown in FIGS. **32-39**. In FIG. **32**, a channel **708** is formed in tibia **712**. A channel **716** is formed in the femur **720**. A guide suture **705** is positioned in aperture **610** of cleat **600** and ends **706** and **707** are threaded through the tibia channel **708** and femur channel **716**. A guide suture **709** is positioned in aperture **612** of cleat **600** and ends **710** and **711** are positioned through the tibia channel **708** and femur channel **716**. The tensioning tail end **622** of the flexible connector **621** is also positioned through the channel **708** and femur channel **716**. The graft **700** is positioned on the adjustable fixation flexible connector portion **688**.

[0131] The guide sutures **705** and **709** are then manipulated by the surgeon to draw the cleat **600** through the tibia channel **708** and femur channel **716**, as shown in FIG. **33**. Tension is placed on tail and **622** as indicated by arrow **732** to draw the adjustable fixation flexible connector portion **688** and the graft **700** toward the cleat **600**. The graft **700** will be drawn through the tibia channel **708** and into the femur channel **716** as shown in FIG. **34**. The ends of sutures **701-704** will extend from the tibia channel **708** and can be secured by a suture anchor **728** and indicated. Remaining portions of the tail end **622** and the securing sutures **701-704** can be trimmed.

[0132] The securing of the cleat **600**, adjustable fixation flexible connector portion **688** and graft **700** is shown in FIG. **35-39**. Tension is applied to the tail end **622** as indicated by arrow **732**. The ascending portion **654** is drawn in the direction indicated by arrow **655** and the descending portion **652** is moved in the direction shown by arrow **653**. The ascending portion **648** is moved in the direction indicated by arrow **649**, and the upwardly facing bend **650** and upwardly facing bend **658** move upward and engage the graft **700** causing the graft to move upward in the direction indicated by arrow **735**. The graft applies a counteracting force against the adjustable fixation flexible connector portion **688** as indicated by arrow **739** and **741**.

[0133] In the initial stages of tensioning, the downwardly facing inner bend **626** and downwardly facing outer **638** are loosely positioned in the oval bore **604** and tapered bore **608**, as indicated in FIG. **36**. As shown in FIG. **36A**, the flexible connector **621** has a radius R_1 and the bridge portion **618** has an arcuate surface **619** providing a tapered radius of R_2 . Increased tension on the tail end **622** is shown in FIG. **37** by arrow **732**. The graft **700** continues to apply a downward force as indicated by arrow **745**. The portion **623** of the flexible connector **621** is forced against the seat portion **618** of cleat **600** and particularly the arcuate surface **619**. The cross-sectional area of the

flexible connector **621** is greater than the cross-sectional area defined by the radius **R2**, and accordingly the portion **623** is wedged into the arc defined by the arcuate surface **619** of the tapered bore **608**. This is shown in FIG. **38**. The downward facing outer bond **638** is drawn downwards through the tapered bore **608** and toward the seat portion **618**, as is the downward facing inner bend portion **626**. The portion **623** is thereby firmly locked in position, and will remain locked when the tail end **622** is removed. The graft **700** is thereby securely positioned in the femur channel **716**, as shown in FIG. **39**.

[0134] There is shown in FIGS. **40-42** a double-sided embodiment **800** of the adjustable fixation device. The double-sided embodiment **800** has a first adjustable fixation device **812** and a second adjustable fixation device **820**. A graft **828** is disposed between and of the first adjustable fixation device **812** and the second adjustable fixation device **820**, and can be secured together by wrapped line or suture **832**. The first adjustable fixation device **812** has a cleat **804** and is formed with a flexible connector **816** that includes a tensioning end. The second adjustable fixation device **820** has a cleat **808** and a flexible connector **824** with a tensioning end. The structure and operation of the first adjustable fixation device **812** and the second adjustable fixation device **820** can be similar to that shown in FIGS. **26-39**, or in one or more of the other embodiments shown herein.

[0135] The installation of the double-sided embodiment **800** is performed between the tibia **836** and femur **844**, as indicated by arrow **852** in FIG. **41**. A first cavity **840** is formed in the head of the tibia **836**. A second cavity **848** is formed in the head of the femur **844**. The double-sided embodiment **800** is inserted between the tibia **836** and femur **844** as shown by arrow **852** and FIG. **41**. Guide sutures (not shown) are used to guide the first cleat **804** through the first cavity **840**, and the second cleat **808** through the second cavity **848**, as shown in FIG. **42**. The graft **828** is thereby positioned within the first cavity **840** and the second cavity **848**. Tension indicated by arrow **826** is applied to the tension end of flexible connector **816**, and tension indicated by arrow **830** is applied to the attention end of flexible connector **824**. The first adjustable fixation device **812** and second adjustable fixation device **820** can thereby be tightened and secured in place to secure the graft **828** in position.

[0136] There is shown in FIGS. **43-46** an alternative embodiment particularly useful for syndesmosis repair. As shown in FIGS. **43-44**, this embodiment **900** of the adjustable fixation device can have a cleat **902** with apertures **904** and **908**. A flexible connector portion **922** has a tail **924** and a portion **934** of the flexible connector can pass through the bore **904** and over a seat portion **918** through the bore **908** to form a downward facing inner bend **936**. A portion of the flexible connector can be captured longitudinally and secured to itself such that the inner bend **936** forms part of a closed loop. A portion **937** passes through the bore **904** forming an upward facing bond **939**. A portion **946** passes through the bore **908** such that a downward facing outer bend portion **938** is over the downward facing inner bend portion **936**. The portion **946** can turn upward at an upward facing bend **958** to provide an ascending portion **952**.

[0137] The ascending portion **952** can pass over the upward facing bend **939** to form a downward facing bend **942** and a descending portion **948**. The portion of the flexible connector forming the downward facing bend **942** is slidable relative to the upward facing bend **939**, and forms and intra-connector sliding loop engagement portion. The descending portion **948** can turn upward at an upward facing bend **950** and form an ascending portion **954**. A portion of the ascending portion **954** is slidably captured longitudinally within the portion **946** and exits as the tail **924**.

[0138] A second cleat **928** has a groove **960** to receive the upward facing bend **958** and a groove **964** to receive the upward facing bend **950**. The groove **960** communicates with apertures **961** and **963**. The groove **964** communicates with apertures **962** and **965**. The portions of the flexible connector corresponding to the upward facing bend **950** and upward facing bond **958** move within the groove and the apertures to provide a sliding interaction of these portions of the flexible connector with the second cleat **928** and form a connector-device sliding loop engagement portion.

[0139] The flexible connector transversely engages the downward facing inner bend **936** and abuts

the bridge portion **918** and is positioned between the bridge portion **918** and a corresponding portion of the downward facing bend **936**. The flexible connector then transversely passes through the downward facing outer bend **938**. The tail **924** will provide tensioning to tension and secure the adjustable fixation device **900** in position.

[0140] A syndesmosis repair is shown in FIGS. **45-46**, where the fibula **970** is secured to the tibia **972** over the talus **974**. Guide holes **905** and **907** can be provided in the cleat **902** and can be used too position guiding sutures for guiding the cleat **902** into proper position. Guide holes **931** and **933** and can be provided in the cleat **928** to receive guiding sutures for guiding the cleat **928** into proper position. The cleat **902** communicates with a plate **978** to provide support for the fibula **970**. A tensioning force is applied to the tensioning end **924** of the flexible connector to draw the second cleat **928** toward the plate **978**.

[0141] While the exemplary embodiments described above have been chosen primarily from the field of arthroscopic surgery, one of skill in the art will appreciate that the principles of the invention are equally well applied, and that the benefits of the present invention are equally well realized in a wide variety of other surgical and medical systems including, for example, conventional surgery. Further, while the invention has been described in detail in connection with the presently preferred embodiments, it should be readily understood that the invention is not limited to such disclosed embodiments. Rather, the invention can be modified to incorporate any number of variations, alterations, substitutions, or equivalent arrangements not heretofore described, but which are commensurate with the spirit and scope of the invention. Accordingly, the invention is not to be seen as limited by the foregoing description, but is only limited by the scope of the appended claims.

Claims

1. An adjustable fixation device, comprising: a cleat comprising first and second cleat openings and a connector seat between the openings; a flexible connector having first and second cleat loops through the first and second openings of the cleat and a tensioning end portion, the connector comprising an intra-connector sliding loop engagement portion and a connector-device sliding loop engagement portion, the tensioning end portion of the flexible connector further comprising an axial sliding connector-connector engagement portion and a transverse sliding connector-connector engagement portion transversely engaging the first and second cleat loops and extending between portions of the cleat loops and the connector seat.
2. The adjustable fixation device of claim 1, wherein the flexible connector is a suture.
3. The adjustable fixation device of claim 1, wherein axial sliding connector-connector engagement portion comprises a portion of the tensioning end being positioned in an axial channel through a portion of a cleat loop.
4. The adjustable fixation device of claim 3, wherein the axial channel is formed by axial threading of the tensioning end through the portion of the cleat loop.
5. The adjustable fixation device of claim 1, wherein the transverse sliding connector-connector engagement portion comprises a portion of the tensioning end being transversely threaded through a cleat loop.
6. The adjustable fixation device of claim 5, wherein the transverse sliding connector-connector engagement portion comprises a portion of the tensioning end being positioned in a transverse channel through a portion of a cleat loop.
7. The adjustable fixation device of claim 6, wherein the transverse channel is formed by transverse threading of the tensioning end through the connector-connector engagement portion of the cleat loop.
8. The adjustable fixation device of claim 1, wherein the cleat further comprises lateral guide holes for receiving guiding sutures.

9. The adjustable fixation device of claim 1, wherein at least one of said first and second cleat openings is tapered.

10. The adjustable fixation device of claim 1, further comprising a protective cover for the flexible connector at a portion of the flexible connector comprising at least one selected from the group consisting of the intra-connector sliding loop engagement portion and the connector-device sliding loop engagement portion.

11. An adjustable fixation system, comprising: an adjustable fixation device, the adjustable fixation device comprising a cleat comprising first and second cleat openings and a connector seat between the openings, and a flexible connector having first and second cleat loops through the first and second openings of the cleat and a tensioning end portion, the connector comprising an intra-connector sliding loop engagement portion and a connector-device sliding loop engagement portion, the tensioning end portion of the flexible connector further comprising an axial sliding connector-connector engagement portion and a transverse sliding connector-connector engagement portion transversely engaging the first and second cleat loops and extending between portions of the cleat loops and the connector seat; and, an implantable medical device slidably connected to the adjustable fixation device at the connector-device sliding loop engagement portion.

12. The adjustable fixation system of claim 11, wherein the implantable device is a graft.

13. The adjustable fixation system of claim 11, wherein the implantable medical device is a second cleat comprising first and second holes, and wherein the connector-device engagement portion of the flexible connector is threaded through one of the holes in a first direction and through the other of the holes in a second direction forming a sliding connector-device engagement.

14. The adjustable fixation system of claim 11, wherein the implantable medical device comprises a plate comprising first and second holes, and wherein the connector-device engagement portion of the flexible connector is threaded through one of the holes in a first direction and through the other of the holes in a second direction forming a sliding connector-device engagement.

15. The adjustable fixation system of claim 11, wherein the cleat further comprises lateral guide holes for receiving guiding sutures.

16. The adjustable fixation system of claim 11, further comprising a second adjustable fixation device having a second connector-device sliding loop engagement portion, the implantable medical device being slidably connected between the connector-device sliding loop engagement portion of the adjustable fixation device and the connector-device sliding loop engagement portion of the second adjustable fixation device.

17. The adjustable fixation system of claim 11, further comprising a protective cover for the flexible connector at a portion of the flexible connector comprising at least one selected from the group consisting of the intra-connector sliding loop engagement portion and the connector-device sliding loop engagement portion.

18. The adjustable fixation system of claim 11, wherein the system provides a 4:1 mechanical advantage.

19. A method for the adjustable fixation of an implantable medical device, comprising the steps of: providing an adjustable fixation device, the adjustable fixation device comprising a cleat comprising first and second cleat openings and a connector seat between the openings, and a flexible connector having first and second cleat loops through the first and second openings of the cleat and a tensioning end portion, the connector comprising an intra-connector sliding loop engagement portion and a connector-device sliding loop engagement portion, the tensioning end portion of the flexible connector further comprising an axial sliding connector-connector engagement portion and a transverse sliding connector-connector engagement portion transversely engaging the first and second cleat loops and extending between portions of the cleat loops and the connector seat; providing an implantable medical device slidably connected to the adjustable fixation device at the connector-device sliding loop engagement portion; positioning the adjustable fixation device and the implantable medical device in the body of the patient; applying a tensioning

force to the tensioning end portion of the flexible connector to tighten and secure the adjustable fixation device and the implantable medical device in position in the body of the patient.

20. The method of claim 19, wherein the flexible connector is a suture.

21. The method of claim 19, wherein axial sliding connector-connector engagement portion comprises a portion of the tensioning end being positioned in an axial channel through a portion of a cleat loop.

22. The method of claim 21, wherein the axial channel is formed by axial threading of the tensioning end through the portion of the cleat loop.

23. The method of claim 21, wherein the transverse sliding connector-connector engagement portion comprises a portion of the tensioning end being transversely threaded through a cleat loop.

24. The method of claim 23, wherein the transverse sliding connector-connector engagement portion comprises a portion of the tensioning end being positioned in a transverse channel through a portion of a cleat loop.

25. The method of claim 24, wherein the transverse channel is formed by transverse threading of the tensioning end through the connector-connector engagement portion of the cleat loop.

26. The method of claim 19, wherein the cleat further comprises lateral guide holes for receiving guiding sutures.

27. The method of claim 19, wherein the implantable medical device is a plate and the fixation is for syndesmosis repair.

28. The method of claim 19, wherein the method is applied to at least one selected from the group consisting of ligamentum teres repair, anterior cruciate ligament repair, and syndesmosis repair.

29. The method of claim 19, wherein the method provides a 4:1 mechanical advantage.

30. An adjustable fixation device comprising: a cleat, said cleat including a body member, said body member having an upper surface and a lower surface, said upper surface being disposed in substantially parallel spaced relation to said lower surface, said cleat having disposed therethrough between said upper and lower surfaces a first symmetric hole and a second tapered hole; and a suture portion, wherein a first region of said suture portion is disposed within said first symmetric hole and a second region of said suture portion is disposed within said tapered hole, said suture being configured to include an adjustable splice.

31. The adjustable fixation device of claim 30, wherein said suture portion further includes a loop portion, said loop portion being disposed coaxially within a sheath, said sheath including first and second tails for fixing a graft adjacent to said loop portion of said suture portion.

32. An adjustable fixation device comprises a cleat portion and a suture portion, said suture portion including a suture tail, said suture tail being configured, when placed in tension, to tighten a loop of said suture portion so as to convey a mechanical advantage from said tail portion to said loop.

33. The adjustable fixation device of claim 32, wherein said mechanical advantage comprises a 4:1 mechanical advantage.

34. The adjustable fixation device of claim 32, wherein said loop portion is disposed on a first side of said cleat portion and said tail portion is disposed on a second opposite side of said cleat portion, whereby said tail may be placed in tension after said cleat portion is installed to support a surgical graft with said loop portion.
